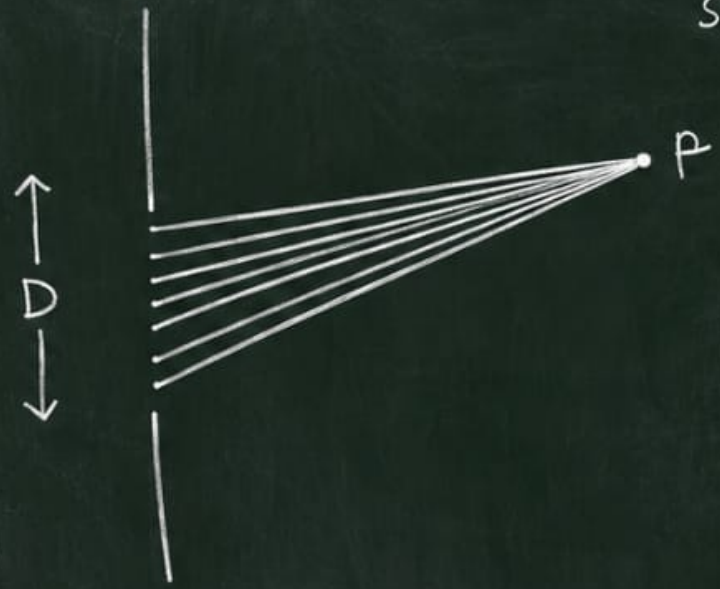
Single slit of width D



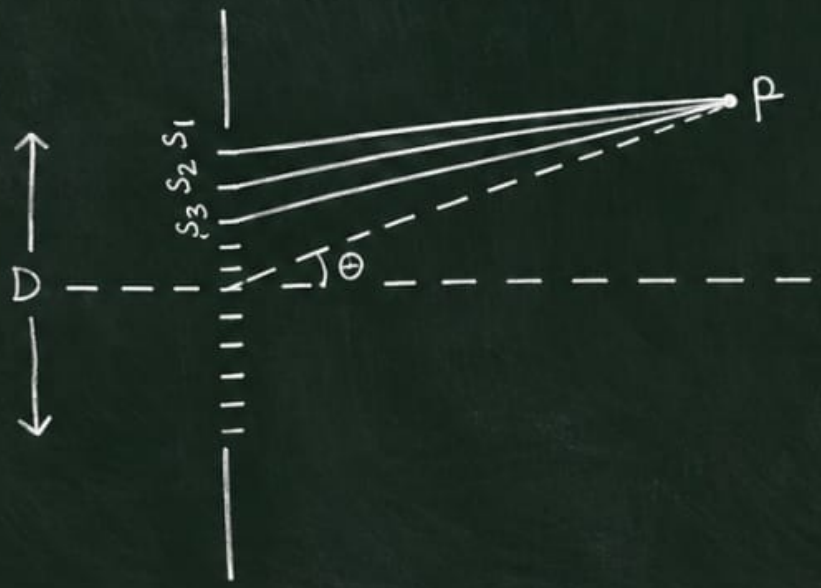
Multiple point sources (infinite no. of them)

Superposition of ## waves from all sources at P

Every bit same as interference, but called diffraction.

How do we add them up? Consider N source (N large)

Observation point P at an angle θ from the centreline.



Pick any 2 sources

Call the path difference

$$S_2P - S_1P = \delta$$

By symmetry, path difference between every consecutive pair is δ. For example

$$S_3P - S_2P = \delta$$

Starting from the bottom of the slit, every consecutive source is off by δ at P.

To find resultant intensity at P, we need to add all these infinite number of waves

$$E_R = A\cos(kr-\omega t) + A\cos(kr-\omega t+k\delta) + \dots + A\cos(kr-\omega t+Nk\delta)$$

$$\left(Nk\delta = D\sin\theta \quad \& \quad N \to^{\text{very large}} \right)$$

$$E_R = R_c \left(A e^{i\phi} + A e^{i\phi} e^{ik\delta} + \cdots + A e^{i\phi} e^{iNk\delta} \right) \quad (3)$$

$$= R_c \; A e^{i\phi} \left(1 + e^{ik\delta} + e^{2ik\delta} + \cdots + e^{iNk\delta} \right)$$

$$S = 1 + q + q^2 + \cdots + q^N$$

$$qS = q + q^2 + \cdots + q^{N+1}$$

$$S - qS = 1 - q^{N+1}$$

$$S = \frac{1 - q^{N+1}}{1 - q}$$

$$E_R = R_c \left[A e^{i\phi} \frac{1 - q^{N+1}}{1 - q} \right] = R_e \left[A e^{i\phi} \frac{1 - e^{ik\delta(N+1)}}{1 - e^{ik\delta}} \right]$$

$$1 - e^{i\alpha} = 2i \, e^{i\frac{\alpha}{2}} \sin\frac{\alpha}{2}$$

$$E_R = R_e\left[A e^{i\phi}\ \frac{\cancel{2l}\, e^{i\frac{k\delta(N+1)}{2}}\ \sin\frac{k\delta(N+1)}{2}}{\cancel{2l}\, e^{i\frac{k\delta}{2}}\ \sin\frac{k\delta}{2}}\right]$$

$$= R_e\left[A e^{i\phi}\ e^{i\frac{k\delta N}{2}}\ \frac{\sin\frac{k\delta(N+1)}{2}}{\sin\frac{k\delta}{2}}\right]$$

$$N k\delta = kD\sin\theta$$

$$E_R = R_e\left[A e^{i\left(\phi+\frac{kD\sin\theta}{2}\right)}\ \frac{\sin\left(\frac{kD\sin\theta}{2}\right)}{\sin\frac{kD\sin\theta}{2N}}\right] \rightarrow R_e\left[NA e^{i\left(\phi+\frac{kD\sin\theta}{2}\right)}\ \frac{\sin\left(\frac{kD\sin\theta}{2}\right)}{\frac{kD\sin\theta}{2}}\right]$$

$$\boxed{|E_R|^2 = I_0 \cdot I_0\ \frac{\sin^2\beta/2}{(\beta/2)^2}\ ,\ \beta = kD\sin\theta}$$

Dark fringes

$$I = 0$$

$$\frac{(\sin \beta/2)^2}{(\beta/2)^2} = 0$$

$$\frac{kD\sin\theta}{2} = \pm\pi, \pm 2\pi \quad \text{(zero not included!)}$$

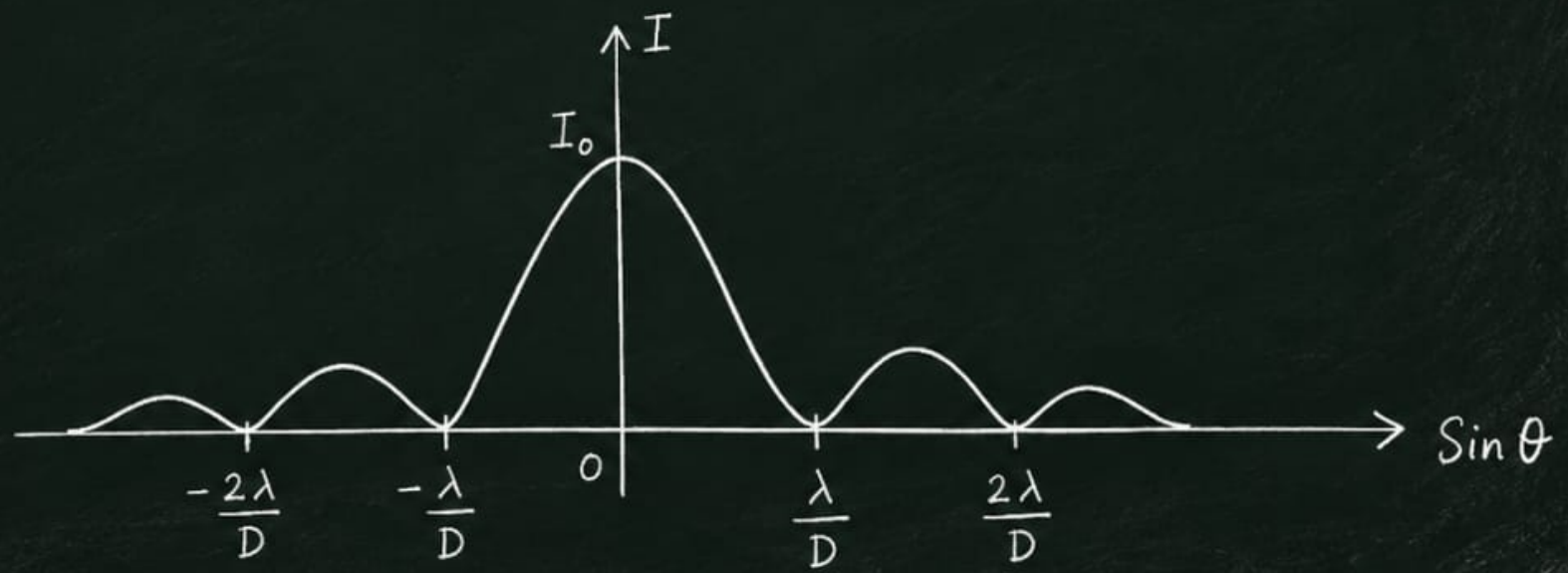
$$D\sin\theta = n\lambda$$

$$n = \pm 1, \pm 2, \cdots$$

$$I = I_0 \frac{\sin^2(\beta/2)}{(\beta/2)^2}$$

$$\beta = \frac{2\pi D}{\lambda} \sin\theta$$

$\sin\theta$	$\beta/2$	$\sin^2 \beta/2$	I
0	0	0	I_0 (L'opital)
$\frac{\lambda}{D}$	π	0	0
$\frac{2\lambda}{D}$	2π	0	0



Finding Maxima is harder

$$I = I_0 \frac{\sin^2(\beta/2)}{(\beta/2)^2}$$

$$\frac{\beta}{2} = \frac{2\pi D}{\lambda}\frac{\sin\theta}{2} = \frac{\pi D}{\lambda}\sin\theta$$

One might think maxima are $\frac{\beta}{2} = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \cdots$

i.e. $\beta = \pm\pi, \pm 3\pi, \cdots$

i.e. $\beta = (2m+1)\pi, \quad m = 0, \cdots$

But there is a $(\beta/2)^2$ in the denominator.

$$\frac{\partial I}{\partial \beta} = 0 \Rightarrow \frac{\partial}{\partial \beta}\left(\frac{\sin^2 \beta/2}{(\beta/2)^2}\right) = 0. \Rightarrow \frac{\beta}{2} = 4.493, 7.725, 10.904, \cdots$$

$$\beta = 2.860\pi, 4.918\pi, \cdots$$

Central Maxima width

$$\sin\theta = \frac{\lambda}{D}$$

Decrease D, width increases.

→ Smaller slit, wider central Maxima